Project data

Call number	2027_BIH
Project title	Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems
Exit Strategy & Impact Assessment (see ? icon for more information)	Exit strategy - Modular framework: each work package yields standalone results; failure of one component does not jeopardize others. - Fallbacks: simpler neural/graph-based scoring if DDMs/GFlowNets underperform; bandit-based control if online adaptation is unstable; LLM part exploratory, non-critical - Two independent applications (MCS, DynEADARP) plus classical benchmarks: if one loses relevance, resources shift to the other - Staff changes: shared code repository, documentation, co-supervision and mutual visits ensure knowledge transfer; each partner can continue its lead tasks independently Impact assessment - Scientific: generative AI embedded in robust metaheuristics is expected to boost effectivity of optimization, which may have strong impact in many applications; - ≥2 joint journal and ≥2 conference papers - Open-source software, training of young researchers, basis for EU follow-up proposals - Applications: efficient shared e-mobility, network data analysis etc.
Monitoring & Indicators for Target Achievement (see ? icon for more information)	Monitoring: - Monthly online meetings of both teams; progress review at two mutual visits per year - Developed components are continuously evaluated on benchmark scenarios and/or instances of the two applications - Shared code repository and benchmark suite; reproducible experiment logs - Annual progress report; switch to fallbacks (simpler scoring models, bandit control) if milestones slip Indicators for Target Achievement: - Methods with learning beat CMSA/LNS baselines and the state of the art on MCS and DynEADARP (objective/optimality gap, runtime, time-to-target, robustness) - Generalization to larger or structurally different instances works - Open-source modular CMSA/LNS framework with learned DDM/GFlowNet/LLM components - Planned publications submitted/accepted - Research visits held, students supervised

Added Value & Internationalisation (see ? icon for more information)	Added value (Austrian research sector & institution): - Strengthens Austrian expertise at the AI–operations research interface (generative AI, LNS, hybrid metaheuristics) and in combinatorial optimization more generally - Access to complementary CMSA, MCS and mathematical-modelling know-how of the BiH partner - Joint high-ranking publications (≥2 journal, ≥2 conference papers) and open-source software - Training of PhD students/early-stage researchers via co-supervision and research stays - Transfer to applications, e.g. sustainable e-mobility (DynEADARP) - Basis for joint Horizon Europe proposals - Expands strategic cooperation with Southeast Europe/Western Balkans - Promotes researcher mobility (two mutual visits per year) and international co-supervision
Application to further programs	No
Planned duration	01-01-2027 - 31-12-2028
Scientific discipline-1	NATURAL SCIENCES; TECHNICAL SCIENCES
Scientific discipline-2	Computer Sciences
Scientific discipline-3	Artificial intelligence; Computational intelligence; Machine learning
Number of persons in home country	4
Number of persons in partner country	5

Home institution(s)

Technische Universität Wien

Home country	Austria
Faculty/Unit	Fakultät für Informatik
Department/Subunit	-

Partner institution(s)

University of Banja Luka

Partner country	Bosnia and Herzegovina
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

Home project coordinator

Assoc.Prof, Günther RAIDL

Gender	Male
Date of birth	24.05.1968
Email	guenther.raidl@tuwien.ac.at
Nationality	Austria
Person is member of staff (employee) at Austrian Higher Education Institution	Yes
Classification of activity at Austrian Higher Education Institution	409875
Home country	Austria
Home institution	Technische Universität Wien
Faculty/Unit	Fakultät für Informatik
Department/Subunit	-

Partner project coordinator

Assoc.Prof, Marko DJUKANOVIC

Gender	Male
Date of birth	20.10.1990
Email	marko.djukanovic@pmf.unibl.org
Nationality	Bosnia and Herzegovina
Person is member of staff (employee) at Austrian Higher Education Institution	No
Home country	Bosnia and Herzegovina
Home institution	University of Banja Luka
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

Home person(s)

Dipl.Ing., Martin WUSTINGER

Gender	Male
Date of birth	26.03.2001
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Person is member of staff (employee) at Austrian Higher Education Institution	Yes
Classification of activity at Austrian Higher Education Institution	409874
Home country	Austria
Home institution	Technische Universität Wien
Faculty/Unit	Fakultät für Informatik
Department/Subunit	-

Dipl.Ing., Maria BRESICH

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Nationality	Austria
Person is member of staff (employee) at Austrian Higher Education Institution	Yes
Classification of activity at Austrian Higher Education Institution	409875
Home country	Austria
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Faculty/Unit	Fakultät für Informatik
Department/Subunit	-

Dipl.Ing., Laurenz TOMANDL

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Person is member of staff (employee) at Austrian Higher Education Institution	Yes
Classification of activity at Austrian Higher Education Institution	409875
Home country	Austria
Home institution	Technische Universität Wien
Faculty/Unit	Fakultät für Informatik
Department/Subunit	-

Partner person(s)

Assoc.Prof, Marko DJUKANOVIC

Gender	Male
Date of birth	20.10.1990
Email	marko.djukanovic@pmf.unibl.org
Nationality	Bosnia and Herzegovina
Person is member of staff (employee) at Austrian Higher Education Institution	No
Home country	Bosnia and Herzegovina
Home institution	University of Banja Luka
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

MSc, Tatjana ZEC

Gender	Female
Date of birth	07.06.1989
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Nationality	Bosnia and Herzegovina
Person is member of staff (employee) at Austrian Higher Education Institution	No
Home country	Bosnia and Herzegovina
Home institution	University of Banja Luka
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

Ass.Prof., Bojan NIKOLIC

Gender	Male
Date of birth	05.06.1980
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Nationality	Bosnia and Herzegovina
Person is member of staff (employee) at Austrian Higher Education Institution	No
Home country	Bosnia and Herzegovina
Home institution	University of Banja Luka
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

MA, Milan PREDOJEVIC

Gender	Male
Date of birth	21.01.1993
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Person is member of staff (employee) at Austrian Higher Education Institution	No
Home country	Bosnia and Herzegovina
Home institution	University of Banja Luka
Faculty/Unit	Faculty of Natural Sciences and Mathematics
Department/Subunit	-

Cost overview

Material cost	0.00 EUR
Travel Expenses	6,600.00 EUR
Total budget	6,600.00 EUR

Abstract to the project proposal
Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems

This project aims to advance the state of the art in combinatorial optimization by integrating modern generative artificial intelligence techniques into established metaheuristic frameworks, particularly Large Neighborhood Search (LNS) and Construct, Merge, Solve and Adapt (CMSA). The research will investigate how generative AI techniques, such as Denoising Diffusion Models (DDMs) and Generative Flow Networks (GflowNets), can enhance solution construction, neighborhood exploration, destroy and repair mechanisms, and automated heuristic design.

A central objective is to develop a modular hybrid optimization framework combining learning-guided search with classical optimization techniques, including exact subproblem solving and Column Generation. The proposed approaches will be systematically evaluated on established graph optimization problems and two structurally more challenging applications: the Minimum Consistent Subset Problem and the Dynamic Electric Autonomous Dial-a-Ride Problem. Particular emphasis will be placed on solution quality, computational efficiency, scalability, and generalization across diverse problem instances.

The project brings together complementary expertise from Austrian and Bosnian-Herzegovinian research teams, fostering methodological innovation and knowledge exchange. Expected outcomes include competitive learning-enhanced optimization algorithms, reusable software components, high-quality joint scientific publications, and a foundation for sustained international research collaboration.

Financing Plan of the S&T Cooperation WTZ 2027-2029

oead
Agency for Education and Internationalisation

Project Titel: Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems

Project Coordinator AT: ao.Univ.-Prof. Dr. Günther Raidl

Partner country: Bosnia and Herzegovina

AT Budget: min € 5.000, max € 10.000

OUTGOING (Austria to partner country)	**Mobility in (year)**	**Name of Austrian team member**	**Funding of travel costs** max. €1.200	**Travel to**	**Funding of accommodation costs:** €120/per working day, €1.440/per month	**estimated duration of stay days/months**	**total costs**
1	2027	Günther Raidl	300	*Bosnia and Herzegovina*	360	*3 working days*	*660*
2	2027	Maria Bresich	600	*Bosnia and Herzegovina*	720	*6 working days*	*1.320*
3	2027	Laurenz Tomandl	300	*Bosnia and Herzegovina*	360	*3 working days*	*660*
4	2027	Martin Wustinger	600	*Bosnia and Herzegovina*	720	*6 working days*	*1.320*
5	2028	Günther Raidl	300	*Bosnia and Herzegovina*	360	*3 working days*	*660*
6	2028	Maria Bresich	300	*Bosnia and Herzegovina*	360	*3 working days*	*660*
8	2028	Martin Wustinger	600	Bosnia and Herzegovina	720	*6 working days*	*1.320*
Sum							**6.600**

Material costs of the project *	**Costs in (year)**	**Type of material**	**total costs**
None			
Sum			**0**

Estimated total costs of the project **						
Total amount requested						**6.600**

* if essential for achieving the project's objectives; consumables

** The **Estimated total costs of the project** must be at least as high as the **Total amount requested**

In case the estimated total costs of the project are higher than the total amount requested - which is assumed - please briefly state who pays the difference to cover the costs (multiple entries possible):

a) the partner university
b) the Austrian partner university from its own resources
c) the project participants privately
d) further funding
e) .....

1.						
Sum						0

Version 20250217

Prospects for Further Cooperation

Several potential avenues for continued cooperation are intended or may be pursued following the completion of the planned project:

1. **Applications for new joint projects.** The partners intend to apply for new research projects, including similar bilateral initiatives, in the research area addressed by the present proposal. This is a highly active field, with new theoretical and practical challenges emerging and being addressed on a regular basis. It is therefore very reasonable to pursue further joint research projects after the completion of the proposed activities.
2. **Workshops.** The partners intend to co-organize at least one workshop after the end of this project associated with a fitting international conference, e.g. GECCO. This activity would promote the research field and attract other researchers at an international level.
3. **Cooperation in master's-level education.** The Mathematics and Informatics study programme is currently preparing to launch its first cohort of a modern master's programme in computer science. Collaboration at the master's level, including teaching contributions from academics at both institutions, could enrich the educational experience of students. Given the strong overlap between the partners' research interests—particularly discrete optimization, algorithms, and graph theory—such cooperation would be especially appropriate. It could also involve joint supervision of master's theses, participation in collaborative projects, and opportunities for students to benefit from the expertise of both institutions.
4. **Institutional partnerships and student mobility.** The partners intend to strengthen links between their universities through joint applications to programmes such as Erasmus+ and Horizon Europe. Such initiatives could enable master's students from both institutions to spend a period at the partner institution, attend courses relevant to their interests, complete examinations, and collect the corresponding credits recognized by their home institution.
5. **Invited research seminars.** Researchers from each institution shall be invited to present their work within the other institution's research seminar series also after this project. Both partners already organize seminars aimed at doctoral students, academic staff, and invited researchers. Reciprocal seminar visits would help strengthen institutional ties and stimulate new ideas for collaborative projects and joint scientific publications.
6. **Development of an international research network and COST Action.** In the near future, and following an expansion of the network of collaborating researchers, the institutions may bring together additional partners to establish an international research team. This would create a foundation for applying to relevant COST calls, potentially with a proposed COST Action focused on generative artificial intelligence in discrete optimization and the application of these methods to real-world problems involving large-scale networks.

0.1 Günther R. Raidl (*1968)



Position

Associate Professor in the Algorithms and Complexity Group, Institute of Logic and Computation

Research Profile

Combinatorial Optimization, Mathematical Programming, Metaheuristics, Hybrid Optimization Techniques, Transport Logistics, Scheduling, Graph Algorithms

Current Teaching

Algorithmen und Datenstrukturen (8 ECTS); Algorithmics (6 ECTS); Heuristic Optimization Techniques (6 ECTS); Propädeutikum für Informatik (3 ECTS)

Supervision in the Last 5 Years

Number of completed bachelor's theses:	15
Number of completed master's theses:	21
Number of doctoral theses (completed and under supervision):	9

Academic Career

Since 2009	Associate Professor, TU Wien
2005-2009	Temporal Full Professorship for Combinatorial Optimization, TU Wien
2003	Habilitation in Practical Computer Science, TU Wien
1995 – 2003	Assistant Professor and Postdoc Researcher, TU Wien
1992 – 1995	Doctorate in Computer Science at TU Wien
1987 – 1992	Masters degree in Computer Science at TU Wien

Five Selected Publications of the Last 5 Years

[1] Maria Bresich, Günther R. Raidl, and Steffen Limmer. "Revisiting Large Neighborhood Search with On-the-Fly ChargingStation Insertion for the Electric Autonomous Dial-a-Ride Problem". In: *ACM Transactions on Evolutionary Learning and Optimization* 6.2 (2026), pp. 1–31. DOI: 10.1145/3779426.

[2] Enrico Iurlano, Tatjana Zec, Marko Djukanovic, and Günther R. Raidl. "Signed Double Roman Domination on Cubic Graphs". In: *Applied Mathematics and Computation* 471 (2024), p. 128612. DOI: 10.1016/j.amc.2024.128612.

[3] Johannes Varga, Günther R. Raidl, Elina Rönnberg, and Tobias Rodemann. "Scheduling Jobs Using Queries to Interactively Learn Human Availability Times". In: *Computers & Operations Research* 167 (2024), p. 106648. DOI: 10.1016/j.cor.2024.106648. URL: https://doi.org/10.1016/j.cor.2024.106648.

[4] Bojan Nikolic, Aleksandar Kartelj, Marko Djukanovic, Milana Grbić, Christian Blum, and Günther Raidl. "Solving the Longest Common Subsequence Problem Concerning Non-Uniform Distributions of Letters in Input Strings". In: *Mathematics* 9.13 (2021), p. 1515. DOI: 10.3390/math9131515.

[5] Christian Blum, Marko Djukanovic, Alberto Santini, Hua Jiang, Chu-Min Li, Felipe Manya, and Günter R. Raidl. "Solving Longest Common Subsequence Problems via a Transformation to the Maximum Clique Problem". In: *Computers & Operations Research* 125.105089 (2020). DOI: 10.1016/j.cor.2020.105089.

Selected Projects of the Last 5 Years

Since 2023	PI, Learning to Solve Dynamic Vehicle Routing Problems (Dynamic VR), individual project, Honda Research Institute Europe
Since 2022	PI, Cooperative Personnel Scheduling, individual project (CoopSched), Honda Research Institute Europe
2016 – 2024	Faculty member, Vienna Graduate School on Computational Optimization (VGSCO), doctoral college, FWF
2018 – 2024	PI, Cooperative Optimization Approaches for Distributing Service Points, individual project, Honda Research Institute Europe

Empowering Young Talent: Success Stories

Jakob Puchinger (Full Professor at Normandie Business School); Markus Leitner (Associate Professor at Vrije Universiteit Amsterdam); Mario Ruthmair (MIP Development Scholar, Gurobi)

Awards and Prizes

2026	Test of Time Paper Award of the INFORMS Journal on Computing (IJOC)
2024	Best Paper Award at GECCO'2024
2008, 2010, 2013, 2022, 2023	Best Paper Awards at EvoCOP
2012	EvoStar Old Croc Award
2009	Best Paper at HIS'2009
2003	Best Paper Award at ADoRO'2003
1999	Best Paper Award at ACM SAC'1999

Service to the TU Wien

Member of Professor Appointment Committees (1999, 2009); Member of Habilitation Commission (2005, 2007, 2009); Board Member of the Center for Informatics and Society (2016–2018); Member of the Fakultätsrat Informatik (2003–2012); Representative for getTUgether (since 2023)

Service to the Scientific Community

Executive Board Member of ÖGOR (2013–2016); co-chair of the ÖGOR Working Group on Metaheuristics Machine Learning Based Methods (since 2025); Co-founder and Steering Committee Member of EvoCOP (since 2004); Associate Editor of INFORMS Journal on Computing (since 2013) and ACM Transactions on Evolutionary Learning and Optimization (since 2019), Editorial Board Member of Algorithms, Engineering Applications of Artificial Intelligence; Co-Chair of the EUROCAST Workshop on Theory and Applications of Metaheuristic Algorithms since 2011; Local Chair of SOCS (2022); Editor in Chief and Program Chair of GECCO 2009; Regular Reviewer for the Research Foundations in Slovenia, Belgium, Switzerland, Ireland, Germany; Program Committee Member and Regular Reviewer of conferences incl. AAAI, ANTS, CPAIOR, GECCO, EvoCOP, EvoStar, ICORES, IJCAI, LOD, MIC, OR, PATAT, PPSN, STACS, VNS

Brief Description of Partner Institution

Faculty of Natural Sciences and Mathematics, University of Banja Luka, Bosnia and Herzegovina

The Faculty of Natural Sciences and Mathematics at the University of Banja Luka was established in 1996 and is located at Mladena Stojanovića 2, 78000 Banja Luka, Republika Srpska, Bosnia and Herzegovina. The Faculty offers eight study programmes, including Mathematics and Informatics, which prepares graduates for careers in mathematics, computer science, and mathematics and informatics education.

The PI and another member of the proposed project team are more specifically affiliated with the Department of Computer and Information Sciences. The Department has four professors and three teaching assistants who contribute to teaching across several courses. The Department is headed by Professor Dragan Matić. The Faculty also maintains cooperation with approximately a dozen companies and organizations, primarily in the information technology sector. The acting Dean is Professor Goran Trbić.

The research of the team members that is most relevant for the proposed project focuses on theoretical and practical aspects of combinatorial optimization problems on graphs, in particular on diverse variants of domination problems, the CMSA metaheuristic with exact subproblem solving, as well as diverse learning-enhanced approaches of CMSA. These aspects excellently augment the expertise of the Austrian team.

Further information: https://pmf.unibl.org/; https://www.unibl.org/sr/clanice/fakulteti/prirodno-matematicki-fakultet

Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems

Research project for Bosnia and Herzegovina–Austria bilateral cooperation

1. Background and motivation

Combinatorial optimization is a cornerstone of modern decision support in logistics, energy systems, telecommunication networks, transportation, and data-driven planning. Many such problems are defined or can be modeled on graphs. Examples of classical graph problems are finding a maximum independent set (MIS), a maximum cut (MaxCut), a densest subgraph of certain size, dominating sets, and diverse vehicle routing problems. These problems are typically NP-hard and can be formulated as large-scale combinatorial search tasks in which the quality of the solution depends on complex structural dependencies and constraints. Additionally, uncertainty or not yet known information in dynamic scenarios often plays important roles in real applications. Exact methods like mathematical programming techniques are powerful and provide quality guarantees but often become intractable on large or dynamic instances. (Meta-)heuristic approaches typically scale better and are in practice often favored, although they usually do not provide quality guarantees and usually rely heavily on hand-crafted operators, parameter tuning, and domain-specific engineering (Blum and Roli, 2003; Glover and Kochenberger, 2003).

Large Neighborhood Search (LNS) and **Construct, Merge, Solve and Adapt** (CMSA) (Blum et al., 2016) are two well-known and widely used metaheuristics, which are known for their flexibility. Both build on the fundamental idea to iteratively derive from the original problem smaller subproblems that are easier to handle by either an exact or heuristic technique. Solutions to these subproblems are then expected to lead to better and better solutions for the original problem. LNS realizes this principle by **destroy-and-repair** operators, which define subproblems by removing solution components or unassigning variables and solve the subproblem of augmenting the solution to a complete feasible one again, respectively. In contrast, CMSA iteratively creates diverse candidate solutions, typically by fast randomized heuristics, collects their components in a **component pool**, and solves a merging subproblem that asks for a best solution just consisting of pool components; after obtaining a solution, the component pool is updated and the process iterated. LNS and CMSA can be seen to be orthogonal approaches in the way they derive subproblems.

Besides classical optimization techniques, machine learning and artificial intelligence approaches are nowadays gaining substantial importance in combinatorial optimization. Data-driven models can identify structural patterns, approximate promising regions of the search space, and support decisions such as which components are promising to remove from or add to candidate solutions, which variables to branch on, how to be well prepared for unknown future aspects in dynamic/stochastic problems, and how to control the search process adaptively. Recent work in learning-enhanced combinatorial optimization is manifold; for an overview see, e.g., (Bengio et al., 2021).

The proposed project builds on these developments by investigating the **integration and combination of advanced generative AI models with LNS and CMSA on selected classes of graph problems**. In particular, **Denoising Diffusion Models** (DDMs), **Generative Flow Networks** (GFlowNets), (Bengio et al., 2023), and **Large Language Models** (LLMs) will be used for (a) creating large sets of meaningful diversified candidate solutions in CMSA, (b) choosing sets of solution components to destroy in LNS, and (c) augmenting partial solutions in the repair step of LNS. Moreover, the project also investigates the **integration of classical Column Generation** (CG) in CMSA and the **hybridization of LNS and CMSA**, in which their principles are combined. Last but not least, learning-based approaches will also be used to adaptively guide the search process in these metaheuristics, for example by selecting most promising operators or adapting strategy

parameters. The project hypothesis is that implicit knowledge encoded in graph-aware neural models and generative architectures can guide the heuristic search at least as or more effectively than purely hand-crafted problem-specific operators. Our recent research on diffusion-based evolutionary search, GFlowNet-based destroy-set generation, and learning-enhanced CMSA provides a strong scientific foundation for this bilateral project (Akbay et al., 2025; Bresich et al., 2026a; Reixach and Blum, 2024; Reixach and Blum, 2025; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

Developed techniques will on the one hand be benchmarked on classical node-selection problems on graphs such as MIS, MaxCut and the densest k-subgraph problem, but on the other hand also be specifically extended for two structurally more challenging applications: the **Minimum Consistent Subset Problem (MCS)** and the **Dynamic Electric Autonomous Dial-a-Ride Problem** (DynEADARP). The project is designed as a genuinely bilateral initiative in which the two partner teams contribute complementary methodological and application expertise. The Austrian partner will primarily provide expertise in LNS, denoising diffusion on graphs, and GFlowNet-based search as well as in vehicle routing, while the Bosnia and Herzegovina partner will primarily contribute expertise in CMSA, graph-based combinatorial optimization, mathematical formulations, benchmark design, and learning-assisted component generation.

2. State of the art and research gap

The field of learning-enhanced optimization has developed rapidly in recent years. Methodological surveys show that machine learning can support branching, solution candidate construction, algorithm configuration, and other decisions within exact and heuristic solvers. Within metaheuristics, data-driven control has been investigated for neighborhood selection, operator scheduling, solution construction, and search-space reduction. In this context, CMSA and LNS are particularly attractive because their performance depends strongly on how promising partial solutions, components, and destroy sets are created/selected (Reixach and Blum, 2024; Reixach and Blum, 2025; Bresich et al., 2026a).

Recent advances in generative artificial intelligence provide more expressive alternatives to classical machine learning techniques. GFlowNets learn stochastic policies for constructing structured objects with probabilities related to their reward. Unlike classical reinforcement learning, which aims to learn a policy for constructing a single structure that maximizes reward, GFlowNets are particularly suited for sampling *diverse* high-reward structures (Bengio et al., 2023). This property is highly relevant for the integration in CMSA and LNS: Both require not just a single strong prediction but a multitude of structurally different promising candidates. GFlowNets are therefore natural tools for generating diverse candidate subsets, partial solutions, and destroy sets in graph-based optimization (Bresich et al., 2026a).

Denoising diffusion models, well known for their impressive results in image and video generation, also have been adapted to combinatorial optimization (Sun and Yang, 2023). Discrete and graph-based diffusion approaches progressively transform noisy structures into high-quality candidate solutions and can be combined with classical search to improve exploration and diversification. Recent work has demonstrated the potential of diffusion-based evolutionary and adaptive search for hard graph problems, including the MIS and a generalization of the domination problem (Salvà Soler and Raidl, 2026; Wustinger et al., 2026). Similarly to GFlowNets, diffusion approaches are able to create large sets of diversified promising candidate structures.

LLMs have recently emerged as a conceptually different tool for combinatorial optimization. Using LLMs directly as optimizers (Yang et al., 2023) works only for small instances. More promising is their use as generators of heuristic code within an evolutionary loop: FunSearch (Romera-Paredes et al., 2024) found new cap set constructions and improved online bin packing heuristics, Evolution of Heuristics (Liu et al., 2024) and ReEvo (Ye et al., 2024) automatically designed heuristics and metaheuristic components for routing, scheduling, and packing problems, and AlphaEvolve (Novikov et al., 2025) improved best-known results for several mathematical and algorithmic problems. Here, the LLM generates candidate heuristics offline, which are then evaluated and selected automatically. Most of this work, however, targets comparatively simple

heuristics; generating specialized components for hybrid metaheuristics such as LNS and CMSA has hardly been investigated.
Despite many recent advances, important more general methodological questions remain open. Existing approaches typically focus on individual interactions between machine learning and optimization, such as branching, construction, neighborhood selection, destroy-set generation, or search-space reduction. The systematic integration of several learned decisions within modular hybrid metaheuristics remains relatively underexplored.

3. Scientific objectives

The project pursues the following objectives:

(i) To develop graph-aware DDM and GFlowNet-based learning mechanisms for diversified candidate solution creation in CMSA, destroy-set generation and solution augmentation in LNS, targeted towards the selected classes of graph problems.
(ii) To investigate the hybridization of LNS and CMSA as well as CG in a joint framework, benefiting from synergy.
(iii) To investigate the use of LLMs as generators of lower-level heuristics, scoring functions, and search-control rules that can be tested and integrated within a reproducible algorithmic framework.
(iv) To validate these methods on benchmark graph problems including the classical MIS, MaxCut, and densest k-subgraph problems as well as the structurally more demanding MCS and the DynEADARP.

The overall goal is to create a hybrid algorithmic pipeline that fuses the robustness of exact subroutines and metaheuristics with the expressiveness and generalization power of modern generative, neural techniques.

4. Proposed methodology and work packages

The project will be organized around a set of interconnected work packages that mirror the scientific agenda of the proposal.

WP1. Modular hybrid optimization framework

This work package will establish a modular optimization framework centered on CMSA- and LNS-based search. Constructive processes, component generation, destroy/repair mechanisms, and reduced-subproblem optimization will be exposed through common interfaces so that learned policies and generative models can be evaluated under comparable conditions. Exact as well as heuristic subproblem solvers will be used as appropriate, depending on the concrete problem and instance characteristics. The aim is not to force all techniques into a single monolithic algorithm, but to create a common experimental backbone for studying how learning can improve the generation and control of reduced search spaces (Akbay et al., 2025; Reixach and Blum, 2024; Reixach and Blum, 2025).

WP2. DDM and GFlowNet models for destroy-set and candidate generation/augmentation

This work package develops DDM and GFlowNet approaches to generate structured destroy sets, candidate subsets, and component pools that are more informative than randomly sampled neighborhoods. For LNS, the learned model will propose destroy sets that target critical vertices, edges, or components capable of triggering strong repair phases. For CMSA, the model will generate candidate components or partial solutions whose inclusion is likely to improve the objective value. For the repair step of LNS, the generative models will be conditioned on the partial solution and will propose completions that serve either directly as repaired solutions or as warm starts and variable fixings for exact repair. Training will be done primarily offline on representative problem instances, using reward signals derived from the objective function for GFlowNets

and dedicated techniques for obtaining meaningful training data from high-quality solutions for DDMs. Generalization from smaller training instances to larger and structurally different test instances will be studied explicitly. A key research challenge is to combine diversity, feasibility, and reward-aware generation in a way that is compatible with exact repair and search intensification. This direction is aligned with recent work on diffusion-based search and GFlowNet-guided neighborhood generation (Bengio et al., 2023; Bresich et al., 2026a; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

WP3. LLMs for lower-level heuristic synthesis

This work package explores LLMs as offline algorithm-design assistants. LLMs will be used to propose candidate lower-level heuristics, repair operators, scoring functions, neighborhood definitions, and search-control rules from a high-level specification. Generated routines will be translated into executable code, automatically filtered by feasibility and safety checks, and evaluated using objective optimization performance. The goal is not to replace the solver or to invoke an LLM at every search iteration, but to create a reproducible human-AI co-design framework in which LLMs propose candidate algorithmic ingredients and the optimization framework selects them according to empirical evidence. This exploratory direction is motivated by recent work on LLM-based algorithm discovery and heuristic evolution (Romera-Paredes et al., 2024; Liu et al., 2024; Yang et al., 2023).

WP4. Application 1: Minimum Consistent Subset Problem (MCS)

The MCS is defined as follows. Let $G=(V,E)$ be a connected graph in which each vertex has one of k colors. A subset $S \subseteq V$ is consistent if, for every vertex $v \in V$, the vertices of S that are closest to v contain at least one vertex having the same color as v. The MCS asks for a consistent subset of minimum cardinality (Banerjee et al., 2018; Dey et al., 2023).

This problem provides a natural model of **representative or prototype selection on colored graphs**. Vertices may represent objects, observations, or agents; graph distances describe relationships or similarity; and colors represent classes or categories. Application contexts include graph-based classification, representative selection, network summarization, and networked data analysis. MCS is especially suitable for learning-enhanced optimization because the relevance of a vertex depends simultaneously on its color, its structural position in the network, shortest-path relations, and the current set of selected representatives.

The project will benchmark exact formulations, CMSA- and LNS-based hybrid methods, and learned component-generation strategies on synthetic and real-world instances and compare them with relevant state-of-the-art approaches from literature.

WP5. Application 2: Dynamic Electric Autonomous Dial-a-Ride Problem (DynEADARP)

The DynEADARP, as defined in Tomandl et al. (2026), concerns a fleet of electric autonomous vehicles by which incoming travel requests of users consisting of pickup and delivery locations as well as time windows shall be fulfilled. In contrast to classical taxi services, users may share rides as long as detours do not get too long. This problem extends the classical dial-a-ride problem in particular by the need to plan suitable charging stops. Moreover, we consider the stochastic dynamic (online) variant in which requests only become available over time and the planning has to be done in a way to be well prepared for the expected future. This problem provides a realistic setting in which learning-based anticipatory decisions may be particularly valuable (Baty et al., 2024). The project will first focus on the simpler static variant, for which the Austrian partner has already developed a leading LNS (Bresich et al., 2026b; Bresich et al., 2026c). Moreover, the group also already proposed a simple reinforcement-learning-based extension of this LNS towards the DynEADARP, which will be used as a baseline (Tomandl et al., 2026). Building on this, the learned destroy-set and repair models will be extended to the dynamic variant, where they are additionally conditioned on information about expected future requests. The two principal application domains deliberately represent

complementary settings. Their joint study will therefore test whether the proposed learning mechanisms transfer across substantially different optimization problem types.

5. Experimental evaluation and risk contingency

The proposed learning-enhanced methods will be evaluated against conventional CMSA and LNS baselines, manually designed operators, exact approaches where computationally feasible, and relevant recent methods from the literature. Performance will be assessed using objective value or optimality gap, runtime, time-to-target, anytime performance, robustness over repeated runs, diversity of generated candidates, reduced-subproblem size, and generalization to larger or structurally different instances. Training, inference, and optimization times will be reported separately in order to assess whether the computational cost of learning is justified and amortized across repeated problem solving.

The modular design limits the scientific risk of individual learning components. If GFlowNet- or diffusion-based generation does not perform well, the same framework will support simpler neural or graph-based scoring models. If online neural adaptation proves unstable or too expensive, lightweight contextual-bandit or adaptive-probability mechanisms will be used. LLM-based heuristic synthesis is explicitly exploratory and is not a critical dependency for successful project completion.

6. Expected results and impact

The project will generate several tangible outputs. First, it will produce a modular hybrid optimization framework that combines classical search with graph-aware learning, generative models, and decision-focused control. Second, it will contribute with better solution approaches for the considered application scenarios, i.e., MCS and DynEADARP. Third, it will create reusable algorithmic modules for destroy-set generation, candidate construction, and adaptive heuristic control. Fourth, it will produce open and reproducible software components aligned with the planned work packages.

The expected scientific impact lies in demonstrating how generative AI and machine learning can be embedded within robust optimization algorithms rather than used only as isolated end-to-end models. The project will generate methodological knowledge on when learned search components provide measurable benefits, how well they generalize, and how their computational cost should be balanced against optimization gains. We plan to publish at least two joint papers in high-ranking journals, such as Computers & Operations Research, and two more joint conference paper contributions, for example at CPAIOR.

7. Bilateral cooperation and division of responsibilities

The project deliberately couples two complementary algorithmic traditions and two distinct application settings. The Austrian partner will lead work on LNS, generative optimization, diffusion and GFlowNet-based search, and DynEADARP. The Bosnia and Herzegovina partner will lead work on CMSA, graph combinatorial optimization, mathematical formulations, MCS, and learning-assisted component generation.

The partners will jointly design the common interfaces between learning and optimization, compare different learning strategies, study cross-problem and out-of-distribution generalization, and develop the LLM-assisted heuristic-synthesis pipeline. Two mutual research visits will be organized per year around concrete joint tasks—algorithm integration, model training, benchmark generation, computational experimentation, and paper preparation—so that mobility directly supports scientific deliverables.

The collaboration will also support co-supervision of graduate students and early-stage researchers, shared software development, joint publications, and preparation of follow-up European proposals at the intersection of operations research and artificial intelligence.

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Project-Relevant Publications of the PI Günther Raidl

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[PROGRAMME]

I. **Brief description of the project tasks of ALL (including unfunded) project members** – Austria and partner country

name	country	academic status		tasks	dates
Günther Raidl	Austria	☐	PhD student	PI on Austrian side, coordination with both teams, general scientific guidance, algorithm development, theoretical aspects, coordination and refinement of research papers, Focus on LNS, diffusion models, primarily responsible for DynEADARP	2007-2008
		☒	researcher		
Maria Bresich	Austria	☒	PhD student	Focus on LNS, GFlowNet approaches, DynEADARP	2007-2008
		☐	researcher		
Laurenz Tomandl	Austria	☒	PhD student	Focus on LNS-CMSA integration, Column Generation, DynEADARP	2007
		☐	researcher		
Martin Wustinger	Austria	☒	PhD student	Focus on diffusion approaches and their integration in LNS/CMSA, DynEADARP	2007-2008
		☐	researcher		
Marko Djukanovic	Bosnia	☐	PhD student	PI on Bosnian side, coordination with both teams, collaboration environment, general scientific guidance, algorithm development, theoretical aspects, coordination and refinement of research papers; focus on CMSA, online learning, alternative learning approaches, primarily responsible for MCS	2007-2008
		☒	researcher		
Tatjana Zec	Bosnia	☒	PhD student	Focus on mathematical/theoretical aspects, in particular for MCS	2007
Milan Predojevic	Bosnia	☒	PhD student	Implementation of framework, CMSA, online learning, in particular for MCS	2007-2008

Bojan Nikolic	Bosnia	☐ ☒	PhD student researcher	Focus on mathematical/theoretical aspects, CMSA, online learning, in particular for MCS	**2007-2008**

II. Please answer the following questions:

1. What scientific, artistic and personal skills and competencies are requirement from the participants to successfully complete this project? (keywords only!)

- Expertise in machine learning, in particular online learning
- Expertise in metaheuristics, in particular mathematical programming
- Expertise in graph theory, optimization problems on graphs

2. Involvement of stakeholders in project implementation. Planned reflection and feedback activities during and after project implementation. (keywords only!)

- Bi-weekly online meetings, publication of joint research papers
- Mutual research visits with joint workshops, progress reviews and feedback rounds at project milestones
- Presentation at international conferences, feedback from the scientific community and practitioners (e.g., mobility providers); final joint review and planning of follow-up cooperation